

STAT 432: Homework 03

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Question 1: Build an explain-to-me skill

I created a short, explicit-use skill whose intent is to explain a homework question without solving it. I saved it as `explain-to-me/SKILL.md`, installed it in the normal Codex skills directory, and invoked it on Question 2.

In the test, the skill explained that Question 2 asks me to generate and standardize one nearly collinear design, hold that design fixed while resampling two response vectors 200 times, compare OLS with two ridge penalties, and interpret coefficient instability and prediction error. It pointed me to the lecture section on putting covariates on a common scale and suggested generating and checking the fixed design as the first step. It did not carry out the calculation or provide the numerical answer, so the test matched the skill's intended boundary.

The supplied validation script could not run because its Python environment lacked the `yaml` module. I diagnosed this from the import traceback, manually verified the required frontmatter, name, and absence of scaffold placeholders, and then completed the installation and behavioral test.

Question 2: Ridge with nearly collinear predictors

(a) Fixed design

Using seed 43231 and the required scaling convention gave

$$\text{cor}(X_1, X_2) = 0.9977, \quad d_{\max} = 14.1415, \quad d_{\min} = 0.4754.$$

The first two columns are nearly the same. Their sum is a well-observed direction, because the columns move together, while their difference is close to zero and is therefore weakly identified. The small singular value records this weak direction. Noise projected onto it is amplified by OLS, so $\hat{\beta}_1 - \hat{\beta}_2$ should vary much more than $\hat{\beta}_1 + \hat{\beta}_2$.

(b) Repeated-response simulation

I held the standardized $\mathbf{X}$ fixed. Within each of 200 repetitions I independently generated $\mathbf{y}$ and $\mathbf{y}^*$, then reused that same realized pair for all three penalties. Entries below are empirical mean (standard deviation).

λ	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_1 + \hat{\beta}_2$	$\hat{\beta}_1 - \hat{\beta}_2$
0	1.496 (1.544)	1.510 (1.549)	3.005 (0.095)	-0.014 (3.091)
0.02	1.487 (0.162)	1.489 (0.166)	2.976 (0.094)	-0.002 (0.314)
0.20	1.368 (0.046)	1.369 (0.047)	2.737 (0.087)	-0.001 (0.034)

λ	Average training MSE	Average test MSE
0	0.9648	1.0625
0.02	0.9747	1.0493
0.20	1.0768	1.1335

(c) Bias–variance interpretation

OLS is approximately unbiased here, but its two individual coefficients have very high variance. Their fluctuations largely cancel in the sum, which is the direction that matters most for fitted values because $X_1 \approx X_2$. At $\lambda = 0.02$, ridge greatly reduces the variance of each coefficient and of their difference, introduces only mild bias in the sum, and slightly improves average test MSE. At $\lambda = 0.2$, variance falls further but the sum is shrunk appreciably below its true value of 3; the resulting bias raises test MSE. Thus coefficients can move dramatically along a nearly null direction while $\mathbf{X}\hat{\beta}$ changes little.

Question 3: What ridge regression shrinks

(a) Normal equation and uniqueness

Differentiating gives

$$\nabla L_\lambda(\beta) = -\frac{1}{n}\mathbf{X}^\top(\tilde{\mathbf{y}} - \mathbf{X}\beta) + \lambda\beta.$$

Setting this equal to zero and multiplying by n yields

$$(\mathbf{X}^\top\mathbf{X} + n\lambda I_p)\hat{\beta}_\lambda = \mathbf{X}^\top\tilde{\mathbf{y}}.$$

For every nonzero $a \in \mathbb{R}^p$,

$$a^\top(\mathbf{X}^\top\mathbf{X} + n\lambda I_p)a = \|\mathbf{X}a\|_2^2 + n\lambda\|a\|_2^2 > 0$$

when $\lambda > 0$. The matrix is positive definite and hence invertible even when $\mathbf{X}$ is rank deficient or $p > n$.

(b) Singular directions and effective degrees of freedom

For the compact SVD $\mathbf{X} = U_r D_r V_r^\top$,

$$\hat{\beta}_\lambda = V_r(D_r^2 + n\lambda I_r)^{-1}D_r U_r^\top \tilde{\mathbf{y}}.$$

Multiplying by $\mathbf{X}$ gives

$$\mathbf{X}\hat{\beta}_\lambda = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + n\lambda} u_j u_j^\top \tilde{\mathbf{y}},$$

so the shrinkage factor is $\rho_j(\lambda) = d_j^2/(d_j^2 + n\lambda)$. Here $n\lambda = 1$, giving

$$\rho_1 = \frac{100}{101} = 0.9901, \quad \rho_2 = \frac{4}{5} = 0.8000, \quad \rho_3 = \frac{0.04}{1.04} = 0.0385.$$

Including the unpenalized intercept,

$$\text{df}_{\text{eff}} = 1 + 0.9901 + 0.8000 + 0.0385 = 2.8286.$$

(c) Strongest shrinkage

The $d_3 = 0.2$ direction receives the strongest shrinkage. Its small singular value provides little information, so the corresponding least-squares fitted component is especially sensitive to response noise; shrinking it reduces prediction variance. If the true mean has a large component in the u_3 direction, however, retaining only about 3.85% of that component creates substantial prediction bias.

Question 4: Ridge regression and optimization

(a) Conditioning

Without ridge, $\kappa(A_0) = 9/0.01 = 900$. Ridge shifts the eigenvalues to 9.09 and 0.10, so

$$\kappa(A_\lambda) = \frac{9.09}{0.10} = 90.9.$$

The objective's contours are still elongated, but much less so: curvature is more similar across directions.

(b) Gradient-descent error

Because the objective is quadratic,

$$\nabla L_\lambda(\beta^{(k)}) = A_\lambda(\beta^{(k)} - \hat{\beta}_\lambda),$$

and therefore

$$e^{(k+1)} = (I - \eta A_\lambda)e^{(k)}.$$

An eigendirection with unpenalized eigenvalue a has ridge eigenvalue $a + \lambda$, so its error is multiplied by $1 - \eta(a + \lambda)$. Every direction contracts when

$$0 < \eta < \frac{2}{M} = \frac{2}{9.09} \approx 0.2200.$$

For $\eta = 1/M$, the contraction factors are 0 in the strong direction and

$$1 - \frac{0.10}{9.09} = 0.9890$$

in the weak direction. Thus the strong-direction error disappears in one step while the weak-direction error decays slowly. For $\eta = 2.1/M$, the strong-direction factor is $1 - 2.1 = -1.1$. Its magnitude exceeds one, so the nonzero component specified in the question oscillates and grows; the method does not converge.

(c) Statistical tuning versus computation

λ defines the estimator and its bias–variance tradeoff. Improving conditioning by increasing λ may speed an algorithm but can add harmful prediction bias. Numerical choices such as the step size should solve a chosen objective accurately, while the penalty should be selected by a prediction criterion such as cross-validation or GCV.

Question 5: Selecting the real-estate penalty

I matched the data and split files by `row_id`, excluded that identifier, and used the supplied 332/82 train–test split. Within each CV fold, I estimated predictor means and population-form scales from only the other nine folds, centered the corresponding training response, fit every candidate penalty, and evaluated the held-out fold. The final test responses were used only after all three tuning rules were fixed.

(a) Ten-fold cross-validation

The minimum mean validation MSE occurred at

$$\lambda_{\min} = 0.01995.$$

Using the sample standard deviation of the ten fold errors divided by $\sqrt{10}$, the one-standard-error cutoff was 102.2076. The largest grid value below this cutoff was

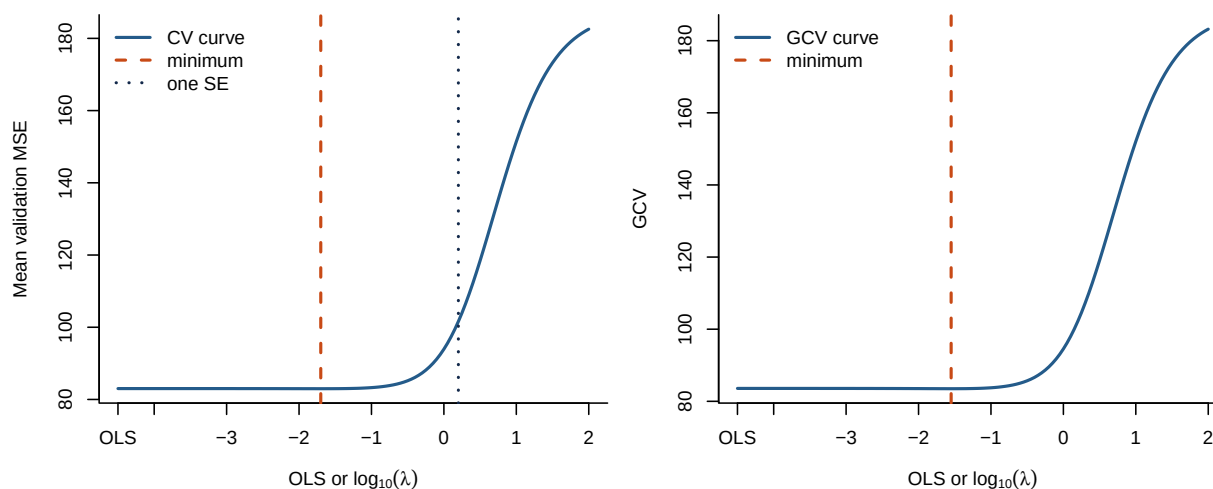
$$\lambda_{1se} = 1.58489.$$

(b) GCV

Using the full training set’s fixed means and scales, I calculated

$$\text{GCV}(\lambda) = \frac{\text{MSE}_{\text{train}}(\lambda)}{\{1 - \text{df}_{\text{eff}}(\lambda)/332\}^2}, \quad \text{df}_{\text{eff}} = 1 + \sum_{j=1}^6 \frac{d_j^2}{d_j^2 + 332\lambda}.$$

The minimum was $\lambda_{\text{GCV}} = 0.02818$.



(c) Frozen-choice final comparison

Rule	λ	Effective df	Training MSE	Test MSE
OLS	0	7.000	80.094	66.322
CV minimum	0.01995	6.773	80.135	66.000
One SE	1.58489	3.033	100.374	81.200
GCV	0.02818	6.689	80.171	65.903

OLS has the smallest training MSE, as expected. CV minimum and GCV select similar mild shrinkage and yield similar test errors. The one-standard-error rule sacrifices several effective degrees of freedom and has higher error on this particular test set. This does not justify switching to GCV after seeing the table: the test set is a final assessment of rules chosen from training information. Selecting the lowest observed test MSE would reuse the test set for tuning and make the reported minimum optimistically biased.

explain-to-me skill

Complete contents of `explain-to-me/SKILL.md`:

```
---
name: explain-to-me
description: Explain a STAT 432 homework question when the student explicitly
  asks to use this skill, without solving the question.
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# Intent

Help me understand a homework question while leaving the mathematical and
computational work for me to complete.

# Instructions

- Read the complete question before responding.
- Explain its main goal, required work, and expected deliverables in clear
  language while preserving its notation.
- For Questions 2 and 3, point to a relevant section of *Ridge Regression:
  Stability Through Shrinkage*. For Questions 4 and 5, use *From a Penalized
  Objective to a Fitted Ridge Model*. Briefly explain why that section helps;
  do not invent a section you cannot verify.
- Suggest one reasonable first step and ask what remains unclear.
- Do not perform calculations or derivations, write solution code, or provide
  a final answer.
```